

INDU 211/2X 2025 – Assignment #2

Problem 1. True or False question

With reference to the following feasible transportation plans, identify if any of the following statements is true or false. If you want, you may add a brief explanation or comment on your answer.

Total cost: $6 \times 15 + 3 \times 20 + 4 \times 21 + 5 \times 21 + 2 \times 30 = \$399 (\times 100)$

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31	21	42	400
Waco	20	21	30	1000
Huntsville	23	20	15	600
Warehouse demand	300	900	800	2000

Total cost: $6 \times 15 + 3 \times 20 + 2 \times 21 + 7 \times 21 + 2 \times 42 = \$423 (\times 100)$

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31	200	21	400
Waco	300	20	700	1000
Huntsville	23	20	600	600
Warehouse demand	300	900	800	2000

- a) The first solution with a total cost of \$39,900 may be optimal
- b) The first solution with a total cost of \$39,900 may not be optimal
- c) The first solution with a total cost of \$39,900 is definitely optimal
- d) The first solution with a total cost of \$39,900 is definitely not optimal
- e) The second solution with a total cost of \$42,300 may be optimal
- f) The second solution with a total cost of \$42,300 is definitely not optimal
- g) An optimal solution of this problem means that any other solutions of the problem will have the same or higher total cost.
- h) The two solutions obtained for this transportation problem are solutions of the “model”, may need further evaluations before they are implemented to actually solve the real-world problem.
- i) It is possible that a general transportation problem model does not have any solutions, depending on the problem conditions and data such as amounts of supply, demands and other requirements.
- j) The unit transportation cost data used to find the solutions are likely to be estimated values and careful evaluations are needed to make sure they are accurate.

Problem 2 (based on Problem 2 in Chapter 4 of textbook)

Following the “lowest cost first” method for solving transportation problems as discuss in *Facility Location Problems*, solve the balanced transportation problem with data given in the following table so that the demand of each store is met by the supplies from the wholesale centers. Calculate the total transportation cost based on the solution you found.

From \ To	Richmond	Philadelphia	Cleveland	Knoxville
Louisville	\$90	\$90	\$100	\$70
Morgantown	\$85	\$70	\$60	\$130

Store:
Demand: Richmond Philadelphia Cleveland Knoxville
 10,000 5,000 5,000 10,000

Wholesale center: Louisville Morgantown
Capacity: 10,000 20,000

Problem 3 (Chapter 5, Problem 1)

1. Fork trucks are located at sites throughout the plant as noted below and are needed at other sites as shown. Formulate as a transportation programming problem and obtain a solution by using the least cost method (see Figure P5.1).

Site where fork trucks are available:

Site	A	B	C	D
Number Available	3	1	2	1

Sites where fork trucks are needed:

Site	X	Y	Z
Number Needed	4	2	1

Distance Matrix (Entries in \$)

From \ To	X	Y	Z
A	5	4	2
B	2	1	2
C	5	4	5
D	3	4	6

Fig. P5.1 Distance matrix (entries in \$).

Problem 4 (based on Chapter 5, Problem 3)

A fork truck is currently located at the tool room, and it must visit the machine shop, paint room, raw materials storage, and finished goods storage. Given the distances in the following figure, formulate as a traveling salesman problem and determine:

1. A solution found by using nearest-neighbor method as discussed in class. Calculate the total travel distance of this solution.
2. A solution found by using “next-nearest-neighbor” method. In using this method, each time the truck will go to the next stop which is the next-nearest neighbor instead of the nearest neighbor of its current position. Calculate the total travel distance of this solution.

From \ To					
	A	B	C	D	E
(A) Tool Room	∞	4	10	8	7
(B) Machine Shop	4	∞	12	7	6
(C) Paint Room	10	12	∞	5	4
(D) Raw Materials Storage	8	7	5	∞	3
(E) Finished Goods Storage	7	6	4	3	∞